

V2 Touch to Switch Effects

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[Feature Description](#)

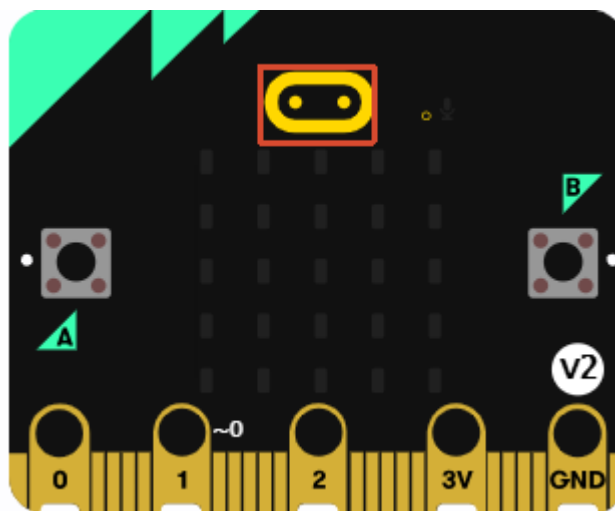
[Experimental Steps](#)

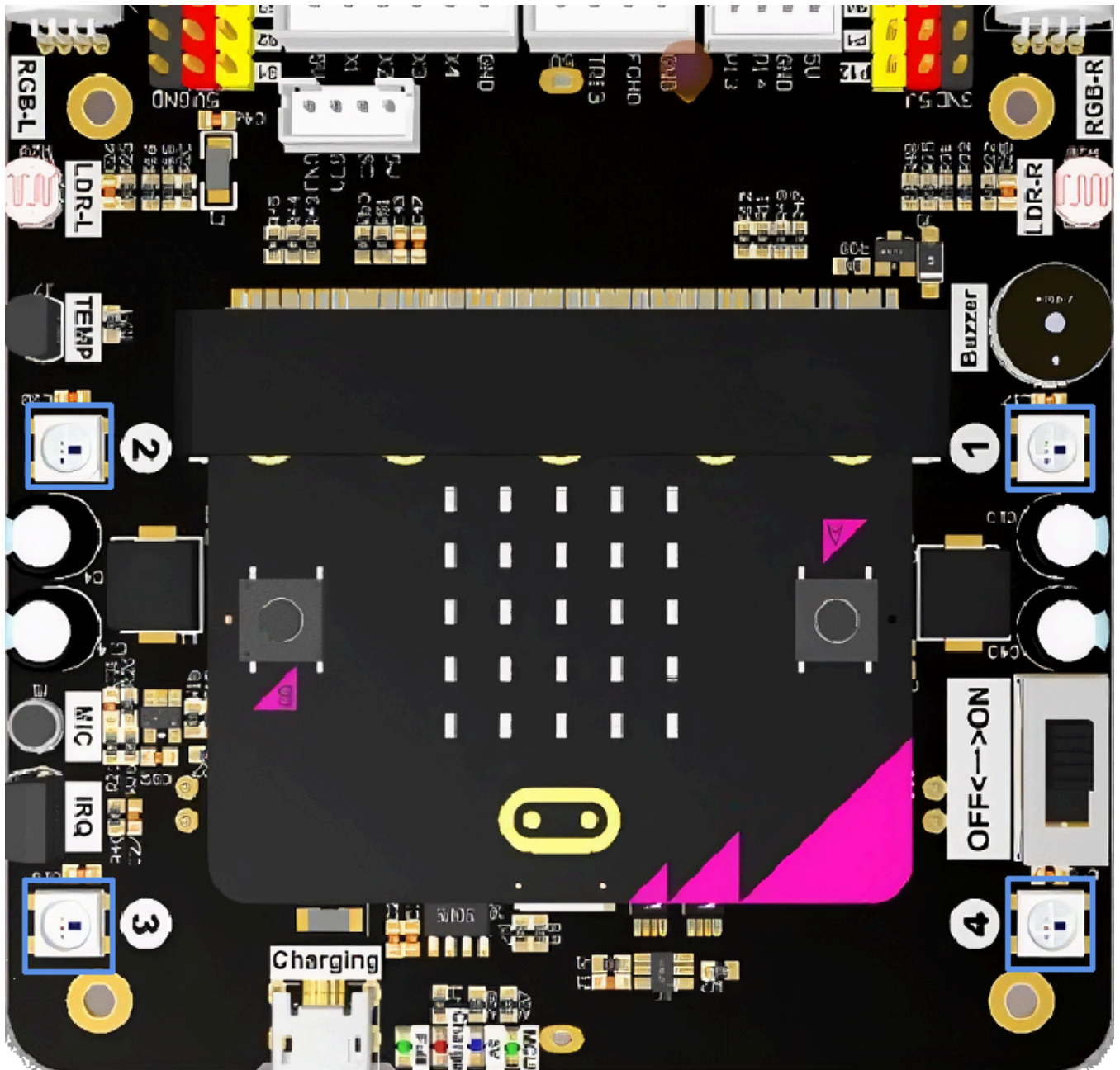
[Code Implementation](#)

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Feature Description

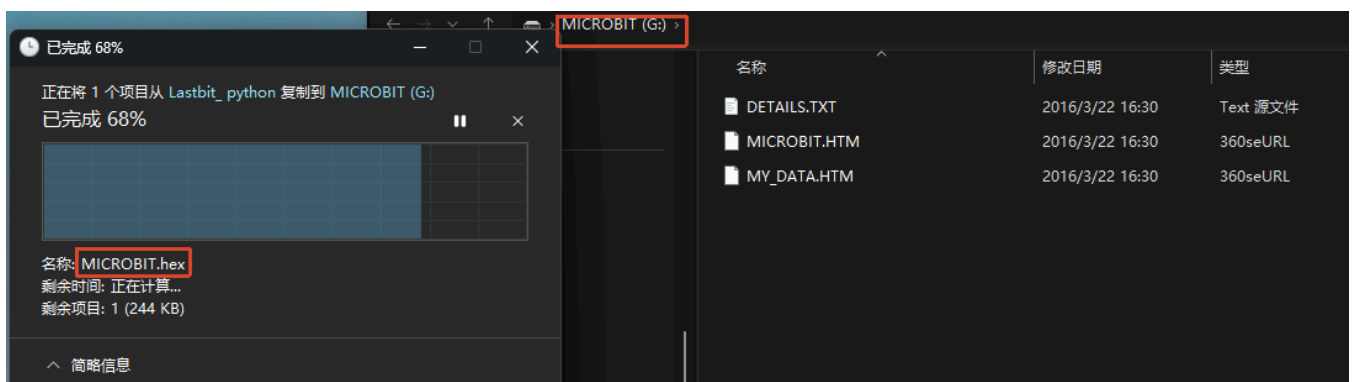
In this section, we will learn to use the touch Logo so that each time you touch it, the color switches to a different one.





Experimental Steps

Before flashing, make sure `LASTBIT.hex` has been flashed; otherwise, an error will occur and the module cannot be imported.



1. Before flashing, switch the switch to the OFF position and connect the computer to the microbit main

board using a microUSB cable. **Note: it should be the Microbit interface, not the expansion board's Charging port.**

2. Open the Mu software. Here you can directly copy the program source code we provide; the operation steps are as follows. You can also enter the code yourself in the editor window. Note! All English characters and symbols should be entered in English input mode. Use the Tab key for indentation, and end the last line with a blank line.

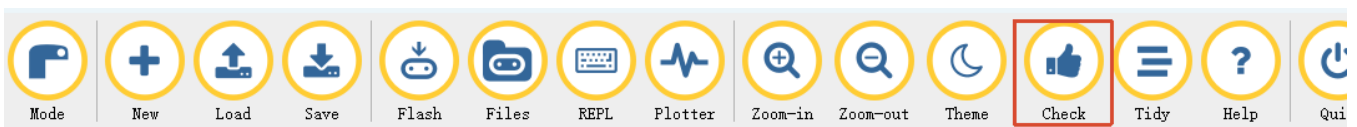
Click the **New** button in the toolbar at the top left, and a file titled untitled will appear.



3. Copy the content of the **Code Implementation** section provided below into the current file. You will notice a small red dot after the title bar, indicating that the code content has been modified but is not yet saved.

Whether or not you save does not affect the code check and flashing.

4. At this point, you can click **Check** to check whether the code format has any problems.



5. If you have connected the MICROBIT to the PC in advance following the steps above and have flashed **LASTBIT.hex**, the next step is to download the current program to the Microbit.

Click the **Flash** tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently, and the editor will show **Copied code onto micro:bit.** at the bottom, indicating that the flashing is complete. After flashing, the indicator light will no longer blink.



6. Unplug the Microusb cable and switch the switch to the ON position.

Code Implementation

```
# 7.Touch_to_Switch_Effects
from microbit import *
from lastbit import LASTBIT

bot = LASTBIT()
bot.all_lights_off()

effect_num = 1
last_touch = False
sleep(1000)

while True:
    # wait for touch event
    touched = pin_logo.is_touched()

    # detected touch event and last time not touched
    if touched and not last_touch:
        effect_num += 1
        if effect_num > 9:
            effect_num = 1

        # set lights based on effect number
        if effect_num == 1:
            # red light
            bot.set_headlight_both('RED')
            bot.set_ws_all('RED')
        elif effect_num == 2:
            # orange light
            bot.set_headlight_both('ORANGE')
            bot.set_ws_all((255, 128, 0))
        elif effect_num == 3:
            # yellow light
            bot.set_headlight_both('YELLOW')
            bot.set_ws_all('YELLOW')
        elif effect_num == 4:
            # green light
            bot.set_headlight_both('GREEN')
            bot.set_ws_all('GREEN')
        elif effect_num == 5:
            # blue light
            bot.set_headlight_both('BLUE')
            bot.set_ws_all('BLUE')
        elif effect_num == 6:
            # cyan light
            bot.set_headlight_both('CYAN')
            bot.set_ws_all('CYAN')
        elif effect_num == 7:
            # purple light
            bot.set_headlight_both('PURPLE')
```

```
        bot.set_ws_all('PURPLE')
    elif effect_num == 8:
        # white light
        bot.set_headlight_both('WHITE')
        bot.set_ws_all('WHITE')
    elif effect_num == 9:
        # close all seven color lights
        bot.set_headlight_both('CLOSE')
        bot.set_ws_all('CLOSE')

    sleep(300)

    last_touch = touched
```

`set_headlight_both(color)` : sets the color of the left and right spotlights.

`set_ws_all(color)` : sets the color of all colorful lights, with 9 preset colors (including off).

Experimental Phenomenon

After the program is flashed, each time the Logo is touched, the colorful lights at the rear of the car will switch to a different color. Nine colors are supported by default, and black is the off state.